

R1: Download + Intro to R & Rstudio

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Introduction to R



Artwork by @allison_horst

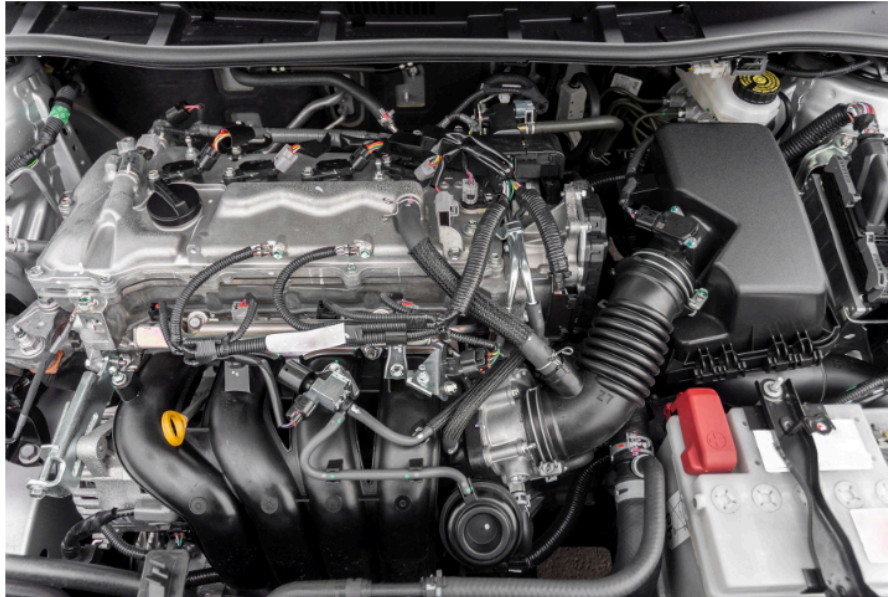
What is R?

- A programming language
- Focus on statistical modeling and data analysis
- Useful for epidemiology, biostatistics, and data science
- Great visualizations



What is RStudio?

R: Engine



RStudio: Dashboard



Modern Dive

- R is a programming language
- RStudio is an integrated development environment (IDE)
 - An interface to use R (with perks!)

We open RStudio on our computer (not R!)

1.1.2 Using R via RStudio

Recall our car analogy from earlier. Much as we don't drive a car by interacting directly with the engine but rather by interacting with elements on the car's dashboard, we won't be using R directly but rather we will use RStudio's interface. After you install R and RStudio on your computer, you'll have two new *programs* (also called *applications*) you can open. We'll always work in RStudio and not in the R application. Figure 1.2 shows what icon you should be clicking on your computer.

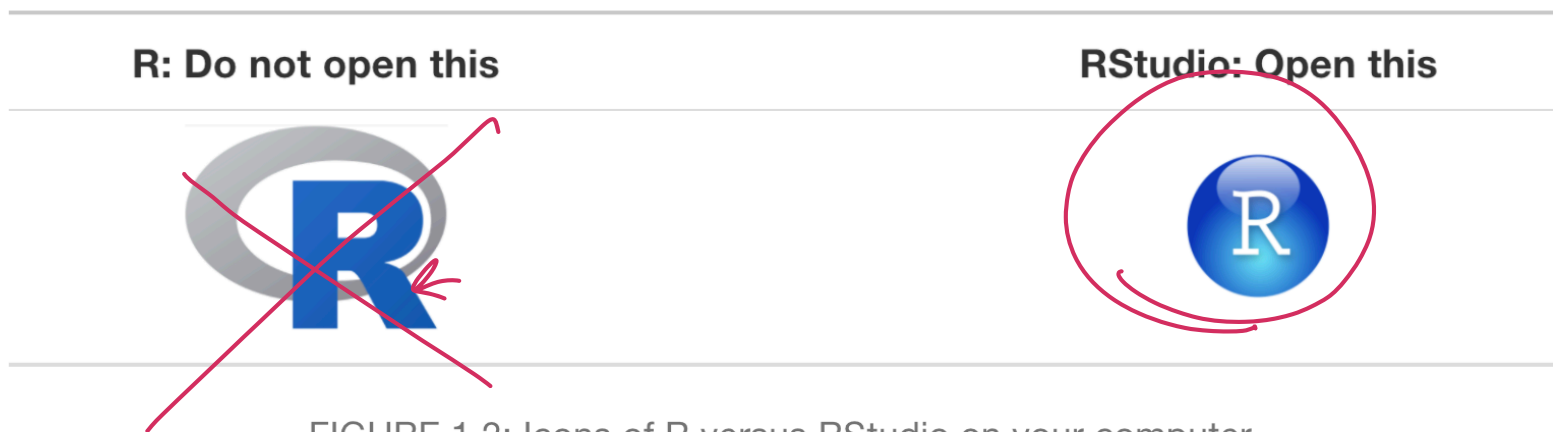


FIGURE 1.2: Icons of R versus RStudio on your computer.

If you have issues downloading on your laptop for in-class use

- There is a really cool online interface that will be sufficient for our in-class use!
- <https://posit.cloud/>
 - If you Google “Posit cloud” it should come up easily
- I don't recommend this for homework use (once we get to it)

So let's take some time to download each!

- Use this link to start: <https://posit.co/download/rstudio-desktop/>
- I will click on this link!
- You must install R first
 - Even if you have R installed already, I highly recommend installing the latest version
 - In the future, you will periodically want to update this
- Install RStudio Desktop Open Source License (second)
- If you get them both installed, then you can open up RStudio and start checking it out

RStudio anatomy

BuzzR

RStudio anatomy

<https://buzzrbeeline.blog/>

Emma Rand

Script file

Write code here
To run code put your cursor on the line and click the **run** button
Edit to correct errors
⇒ record of commands that worked
Save scripts with the **.R** extension
⇒ syntax will be highlighted
⇒ good practice
<- is the assignment operator
⇒ puts what is on the right in to the object on the left
⇒ Assign results if you want to use them again

Console

When you click run, code is sent to the console and executed
> is the prompt
⇒ do not type it
⇒ appears when R is ready for next command
Command output goes here by default
⇒ output is in a different colour
⇒ [1] indicates 3.4 is the first element of the output
⇒ many commands will not have output, the prompt just reappears

Script: where you write code

Console: where output goes

Environment

Name objects by assignment to use them again
All the objects you created in your session
Saving the environment saves all the objects, but not the code with a **.RData** extension
History
A history of every command you sent to the console, mistakes included.
File can be saved but usually you just need the script

Environment: where saved output goes

Packages

Many functions come with R
A huge amount of extra functionality is available in packages
Packages can be installed by clicking the Install button
Help
Access to manual pages for all installed packages
Plots
Figure output appears here

